

ChanYung Lyu

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EDUCATION

Mar. 2019 ~ Feb. 2026	Kyungsung University 메카트로닉스공학과	Busan, Korea
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Thesis: -
Advisor: 제우성, WooSeong Che

BE - Bachelor of Engineering
GPA: 4.052 / 4.5

COURSE PROJECTS

Security Door System Utilizing BLE and Wi-Fi

Role: Personal Project

Tool: Arduino IDE, CATIA, Autodesk Fusion, MIT App Inventor, PyCharm

- Project for IoT Wired and Wireless Communication Lab course of 2025-Spring semester
- Developed a secure door system integrating BLE-based proximity detection and Wi-Fi-based access logging.
- Demonstrated the practical application of smart key technology by reproducing its fundamental features.

Design of an Automated Sorting System Using a Manipulator and QR Codes

Role: Team Leader, Assisted in Hardware Design, Presenter

Tool: Autodesk Fusion

- Project for Comprehensive design course of 2025-Spring semester
- Designed and implemented an automated sorting system that integrates a 4-DOF manipulator with QR code-based machine vision for object recognition.
- Demonstrated a functional prototype that combines kinematics, hardware design, and vision algorithms into a unified mechatronic system

Design of Automatic Door System Using Encoder Motor and PI Control

Role: Personal Project

Tool: Arduino IDE, Code Composer Studio

- Project for Motor Control Engineering course of 2024-Fall semester
- Developed an automatic door system using an encoder DC motor with PI-based closed-loop control for precise position and speed regulation.
- Ensured operational stability and user safety by applying feedback control adaptable to varying loads and external disturbances.

Design of a Door Access and Intercom System Using MSPM0G3507 and Arduino Mega

Role: Personal Project

Tool: Arduino IDE, Code Composer Studio

- Project for Microprocessor Applications course of 2024-Fall semester
- Implemented a door access and intercom system integrating an MSPM0G3507 microprocessor for motor and sensor control with an Arduino Mega for user interface, coordinated via UART communication.
- Demonstrated practical microprocessor applications by combining hardware control and user interaction into a unified system.

EXTRACURRICULAR ACTIVITY

The 15th KSME Student Creative Design Competition

(제 15회 전국학생설계경진대회)

Role: Hardware Design & Prototyping

Tool: Autodesk Fusion, Ansys Workbench

May. 2025 - Sep. 2025

- Developed a hexacopter-based firefighting drone with real-time fire detection and extinguishing functions.
- Implemented YOLOv8-based fire recognition on Jetson Orin Nano Super, integrated with the drone's control system.

The 3rd Future Automotive Autonomous Driving Software Competition

(제 3회 미래형자동차 자율주행 SW 경진대회)

Role: Team Leader, Assisted in Hardware Design, and Administrative Coordinator

Tool: CATIA

Jun. 2024 - Aug. 2024

- Designed and built an autonomous vehicle using provided components within competition regulations.
- Implemented machine vision for traffic light and lane recognition to enable autonomous navigation.
- Applied PID control for smooth steering and LiDAR sensing for rear obstacle detection and parking control.

Asian Student's Capstone Design Contest 2024

(캡스톤디자인 경진대회)

Role: Hardware Design & Prototyping

Tool: CATIA

May. 2024 - Jul. 2024

- Developed an AI-based service robot with autonomous driving, guidance, security, and emotional interaction.
- Designed the system architecture integrating CPU, microcontrollers, LiDAR, cameras, and multiple sensors with Bluetooth communication.
- Implemented algorithms for self-driving, obstacle avoidance, and face recognition using OpenAI-based code.

AWARDS AND HONORS

AWARDS

2024.08 Design Award, The 3rd Future Automotive Autonomous Driving Software Competition

HONORS

2025.03 Academic Scholarship, Kyungsoong University

2024.09 Academic Excellence Scholarship, Kyungsoong University

2024.03 Academic Excellence Scholarship, Kyungsoong University

SKILLS AND TECHNIQUES

English Proficiency

- 2023.07 - TOEIC 735